
MA 323: Applied Databases Practicum

Course Code	MA 323
Course Name	Applied Databases Practicum
L-T-P-C	0-0-3-2
Prerequisites	IC152, MA120
Intended for	B.Tech Maths and Computing
Distribution	Core
Mutual Exclusion	CS 207
Approval	59th BoA (Apr 2025)

Course Contents

- **A few lab lectures** (8 hours spread over the semester):
 - Architectures of DB applications: Client-server; VI-Business logic-DBMS;
 - Browser UI-Web server-Business logic-DBMS.
 - Introduction to SQL
 - Introduction to E-R modelling and MVC
 - Transactions what, how and when?
 - Introduction to a scripting language, eg PHP, Python
 - Introduction to HTML and especially HTML5
 - Introduction to Javascript and Ajax/Comet
 - Introduction to NoSQL
- Lab assignments (listed below) require 3 hours in the lab, preceded by at least 3 hours at home. The weekly assignments are stage-wise demonstrations of the evolution of a mini-project stages are as follows:
 - Week 1-2 Choice of mini-project a useful web-based tool.
 - Week 3-4 Designing the data model and table schemas, testing the tables manually.
 - Week 5-6 Design the UI flow the user view.
 - Week 7 Creating the GUI forms and reports.
 - Week 8-9 Putting together the Web UI flow with the appropriate data access.
 - Week 10 Basic tool ready with full functionality
 - Week 11 Improving UT using javascript and HTML5 features .
 - Week 12 Using AJAX for better user-interaction.
 - Week 13 Substituting portions of the data model using NoSQL databases

Text Books

1. This course will use web-resources to cover course topics.